

The following questions are intended to guide you through the most important concepts that you must understand from the lectures and the material provided.

A good exercise would be to study the lectures/course material until you are able to answer them without having to read or search for it in the slides.

Lecture 4: Limitations, unsolved problems, acceptance, design issues

Questions: factors affecting the performance of biometric systems

- Understand the different factors that may affect, and the three “big” groups into which they can be classified: the biometric trait itself, the system, and the application (slide 11)

Understand the issues and difficulties behind each factor, and to which error they may affect:

- Distinctiveness: limit, inter-user similarity (increases FAR)
- Universality: increases FTE
- Permanence: not totally documented, increases FRR
- Sensor
- User: intra-user variability (increases FRR)
- Environment: indoor, outdoor, etc., can increase both FAR and FRR

Questions: acceptance and privacy (slides 42-46)

- Which factors may affect to better acceptance: comfort, contact, cooperativity needed...
- Other factors: additional information revealed in biometric signals, tracing across databases..
- Smart-cards: how do they work, how they can help towards acceptance/security

Questions: design considerations (slides 47-49)

- Which factors do we have to consider when designing a biometric system
- Which particularities are involved in forensic systems using biometrics